Roberto Gallego Barbarán

Junior Software Engineer | Applied AI & Backend
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PROFESSIONAL SUMMARY

Software Engineering student (University of Málaga, expected 2027) and AI & Automation intern at DEKRA, focused on applied AI and backend. Build LLM agents and RAG in Python (LangChain, LangGraph) and services in Go and Java. 4th place at Hackathon UMA 2026 (Betsson challenge).

TECHNICAL SKILLS

Languages: Python, Go, Java 17, TypeScript, JavaScript, SQL.
Applied AI: LangChain, LangGraph, MCP, RAG, smolagents, LLM agents, computer vision/OCR.
Frameworks: Spring Boot, Spring MVC, Spring Security, JPA, Daml, Canton, gRPC.
Databases: MariaDB/MySQL, vector databases.
Cloud & infrastructure: Docker, Docker Compose, Nginx, Linux, Hugging Face Spaces.
Tools & practices: Git, GitHub Actions (CI/CD), Maven, Playwright, pytest, ruff, JUnit, Mockito.

PROFESSIONAL EXPERIENCE

DEKRA Sept. 2026 – Mar. 2027
AI & Automation Engineer Intern Málaga, Spain

- Build AI agents in Python with LangChain and LangGraph to automate the analysis of laboratory results.
- Implement RAG over vector databases to query technical documentation.
- Connect LLMs to external tools through MCP and orchestrate decision flows with LangGraph state graphs.

IOBuilders (University-Industry Collaboration) 2026
Backend & Blockchain Developer Málaga, Spain

- Wrote Daml contracts for the tokenized bond lifecycle (issuance, cross-participant transfers, allocations and burns) on a Dockerized multi-participant Canton network.
- Built the Go API over the Canton Ledger APIs (gRPC, JSON Ledger API v2), mapping REST endpoints to Daml choices and handling two-step transfers.
- Set up the multi-service environment with Docker Compose and Nginx, and validated the core flows with Playwright end-to-end tests.

PROJECTS

Cava Agent | *Python, smolagents, Gradio, pytest, GitHub Actions* | GitHub 2026

- Turned a fixed LLM pipeline into a ReAct agent (smolagents) that picks its own tools (Yahoo Finance, SMA50/SMA200 trend, news, weather) with a swappable LLM (Groq, Gemini).
- CI/CD in GitHub Actions: ruff and pytest on Python 3.10–3.12, deploying the Gradio demo to Hugging Face Spaces after CI passes; daily Telegram bulletin via cron.

Intelligent Game Validation | *TypeScript, Node.js, Playwright, OCR, GitHub Actions* | GitHub 2026

- 4th place at Hackathon UMA 2026 (Betsson challenge, team of four) with a TypeScript/Playwright tool that validates casino games in cross-origin canvas iframes as the player sees them, with no DOM access.
- Computer-vision pipeline (pixelmatch, template matching, tesseract.js OCR) cross-checked with provider traffic; CI in GitHub Actions with 42 unit tests and a fault-injection negative control.

Bancosol Campaigns | *Java 17, Spring Boot, JPA, Spring Security, JWT, MariaDB* | GitHub 2026

- Implemented the Captain role module of a team Spring Boot web app: dashboard for assigned stores, shifts and volunteers, incident registration, and access limited to each user's own data with Spring Security and JWT.

EDUCATION

University of Málaga 2023 – 2027
BSc in Software Engineering Málaga, Spain

University Extension in Blockchain Technologies (490h) Feb. 2026 – Jun. 2026
University of Málaga (UMA) Málaga, Spain

- Distributed systems, smart contracts (Solidity, Daml), decentralized infrastructure (Canton, gRPC) and blockchain audit fundamentals.

LANGUAGES

Spanish (native), English (C1, Cambridge).